

# Recommended Helium Fill Levels

## 4K Systems

| General information                        |              | Information mainly concerning System Installation |                                                     |                              |                                      | Information mainly concerning CSEs               |                |
|--------------------------------------------|--------------|---------------------------------------------------|-----------------------------------------------------|------------------------------|--------------------------------------|--------------------------------------------------|----------------|
| System type (magnet type)                  | Alarm Level  | He Level for operation / hand-over                | Recommended He Level for start up (before 1st ramp) | Minimum He level for ramping | He consumption for shimming/ ramping | Fill Bandwidth                                   |                |
|                                            |              |                                                   |                                                     |                              |                                      | Target Refill Point (warning level)              | Top Fill Level |
| Aera (OR98)                                | 35%          | 50%                                               | 60%                                                 | 40%                          | 10% **                               | 45%                                              | 65%            |
| Amira, Sempra (AX 93) ****                 | 30%          | 53%                                               | 66%                                                 | 30%                          | 13%                                  | 40%                                              | 60%            |
| Avanto, Avanto fit (OR105)                 | 40%          | 55%                                               | 65%                                                 | 40%                          | 10% **                               | 50%                                              | 70%            |
| Avanto (VB19A SH09; VB19B SH01; and later) | 30%          | 45%                                               | 55%                                                 | 30%                          | 10% **                               | 40%                                              | 60%            |
| Esprea (OR122)                             | 40%          | 55%                                               | 65%                                                 | 40%                          | 10%                                  | 50%                                              | 70%            |
| Esprea (VB19A SH09; VB19B SH01; and later) | 33%          | 50%                                               | 60%                                                 | 33%                          | 10%                                  | 45%                                              | 65%            |
| Essenza (AX60)                             | 40%          | 55%                                               | 65%                                                 | 40%                          | 10% *                                | 50%                                              | 70%            |
| Prisma (OR65)                              | 40%          | 60%                                               | 90%                                                 | 60%                          | 30% **                               | 50%                                              | 70%            |
| Skyra (OR99)                               | 35%          | 60%                                               | 70%                                                 | 60%                          | 10% *                                | 45%                                              | 70%            |
| Skyra/ Spectra (OR 103)                    | 35%          | 50%                                               | 65%                                                 | 40%                          | 15% **                               | 45%                                              | 65%            |
| <b>Sola (OR124)</b>                        | <b>230 L</b> | <b>450 L</b>                                      | <b>645 L</b>                                        | <b>230 L</b>                 | <b>195 L</b>                         | <b>300 L default *****<br/>350 L recommended</b> | <b>600 L</b>   |
| Verio, Skyra fit, mMR (OR97)***            | 50%          | 65%                                               | 80%                                                 | 60%                          | 15% *                                | 60%                                              | 80%            |
| Vida (OR125)                               | 200 L        | 425 L                                             | 645 L                                               | 200 L                        | 220 L *                              | 325 L                                            | 575 L          |
| Terra (OR107)                              | 60%          | 75%                                               | 90%                                                 | 65%                          | 15%                                  | 70%                                              | n.a.           |

## 10K Systems

| General information            |             | Information mainly concerning System Installation |                                                     |                              |                                      | Information mainly concerning CSEs  |                |
|--------------------------------|-------------|---------------------------------------------------|-----------------------------------------------------|------------------------------|--------------------------------------|-------------------------------------|----------------|
| System type (magnet type)      | Alarm Level | He Level for operation / hand-over                | Recommended He Level for start up (before 1st ramp) | Minimum He level for ramping | He consumption for shimming/ ramping | Fill Bandwidth                      |                |
|                                |             |                                                   |                                                     |                              |                                      | Target Refill Point (warning level) | Top Fill Level |
| Allegra (OR92)                 | 30%         | Not applicable                                    | 80%                                                 | 50%                          | 35%                                  |                                     |                |
| Harmony (OR24)                 | 30%         |                                                   | 70%                                                 | 50%                          | 35%                                  |                                     |                |
| Impact (OR41)                  | 30%         |                                                   | 70%                                                 | 50%                          | 35%                                  |                                     |                |
| IMP/HAR, IMP/EXP/HAR (OR42)    | 30%         |                                                   | 70%                                                 | 50%                          | 35%                                  |                                     |                |
| Symphony, SaTS (OR70)          | 35%         |                                                   | 70%                                                 | 50%                          | 35%                                  |                                     |                |
| Prisma fit, Trio, TaTS (OR64)  | 35%         |                                                   | 70%                                                 | 50%                          | 35%                                  |                                     |                |
| Vision, Sym-Vision upgr (OR35) | 30%         |                                                   | 60%                                                 | 50%                          | 35%                                  |                                     |                |

Changes marked in Purple

\* by average of 2 shim iteration

\*\* by average of 3 shim iteration

\*\*\* note the first few magnets have a Alarm level setting of 60%. If this is the case all the min/target levels need to be increased by 10%. To check, set warning level to 60%. If this is not possible your system is one of these few

\*\*\*\* for systems with a SW older than VE11N SP02 the Alarm Level is 15% higher. The recommended He levels need to be adjusted accordingly

\*\*\*\*\* default setting of warning level does not ensure sufficient reaction time until alarm. It is recommended to increase the warning level to 350 L; the next software version will be corrected.